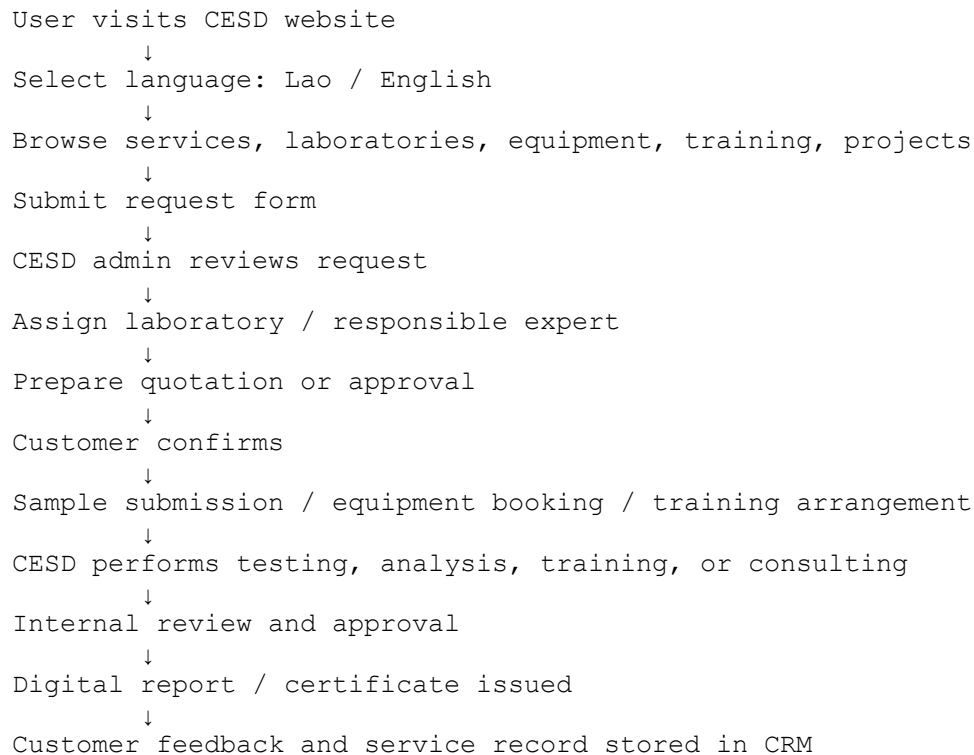


Proposed CESD Service Workflow



Main Workflow Types

1. Testing / Analysis Service

Customer → Request Testing → Admin Review → Lab Assignment → Quotation → Payment/Confirmation → Testing → Report Approval → Report Delivery

This fits CESD's role in measurement, analysis, product inspection, performance testing, and external technical services.

2. Equipment / Laboratory Booking

NUOL User → Login → Search Equipment → Check Availability → Submit Booking → Lab Approval → Use Equipment → Usage Record

This supports student projects, curriculum integration, and faculty research.

3. Training / Workshop Service

Company/User → Request Training → CESD Review → Course Proposal → Schedule Confirmation → Training Delivery → Certificate Issued

This matches CESD's mission for talent development, customized training, and industry collaboration.

4. Joint R&D / Partnership

Partner → Submit Collaboration Interest → CESD Review → Expert Team Meeting → MoU/Agreement → Project Implementation → Output Publication/Showcase

Recommended System Modules

- Bilingual Content Management: Lao / English
- Laboratory & Equipment Catalog
- Service Request Management
- Booking Calendar
- Quotation Management
- Customer/Partner CRM
- Training Management
- Digital Report & Certificate Management
- News / Events / Projects
- Admin Dashboard

Next step: I can prepare the **full Laravel/Next.js functional requirement document and database table structure** for CESD.